Fiscal Year:	FY 2012	**Task Last Updated:**	FY 07/06/2012
PI Name:	Pluth, Janice Ph.D.		
Project Title:	Elucidating the Relationship Between the Effects of Various Radiation Qualities and Cancer Development Processes using Novel Flow-based Assays		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	89154-9900	**Congressional District:**	1
Comments:	NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	09/08/2008	**End Date:**	09/30/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Project will continue as new award; end date is now 9/30/2013, per PI and LBL info (Ed., 7/24/13) NOTE: Extended to 12/31/2014 per PI and LBL info (Ed., 7/12/13) NOTE: Extended to 7/1/2013 per PI and NASA info (Ed., 1/8/2013) NOTE: Extended to 12/31/2012, per C. Guidry/JSC (Ed., 6/11/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ08HD70I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Current radiation quality factors used for determining cancer risk for space radiation do not provide the accuracy needed, and there is debate as to the relevance of the experimental studies they're based upon. Therefore, one of NASA's primary goals is to improve the understanding of the carcinogenic process related to high LET exposures to better define the risk. Recently our current understanding of carcinogenesis has improved, and a number of cancer hallmarks have been identified. Some hallmarks, or surrogate cancer endpoints include genetic instability, telomere length changes and aberrant cellular signaling. Identifying relationships between early and late cell signaling profiles, later surrogate cancer endpoints, and radiation qualities will aid in determining a better risk assessment for cancer formation. In addition, using relevant cell types, such as human epithelial cells in which the majority of cancers develop is also key. In this proposal we will study critical phosphorylation signaling at early and late time points as well as surrogate cancer endpoints of telomere length changes and genomic stability as measured by centrosome amplification, at these same late times post exposure. Epithelial and fibroblast cells from the same individuals will be used to better understand the etiology of cancer, and why certain cell types more easily become carcinogenic. We will also perform studies to understand the heterogeneous nature of high LET radiation. Energy deposition from high LET ions is dense in the track cores, resulting in closely spaced and complex damages, and delta rays originating from cores provide additional exposures similar to X-rays. Using carefully chosen charge, energy and LETs of particles to obtain differing populations of cells hit by tracks and delta rays alone we will investigate how phosphorylation profiles change. These studies will allow us to better quantify the effects of space radiation and its relationship to the carcinogenic process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In this work we are studying critical phosphorylation signaling at early (< 24 h) and late time points (days-weeks) following exposure. In addition, we will determine the relationship between phospho-protein kinetics, and surrogate cancer endpoints of telomere shortening and centrosome amplification. Breast cancer studies have revealed that telomere length decreases during progression from usual ductal hyperplasia (UDH) to invasive cancer, thus telomere length changes can provide a read-out of the carcinogenic process. A second marker we will be following after exposure is centrosome amplification that can lead to gains and losses of whole chromosomes and is believed to be due to continuous chromosome mis-segregation during mitosis resulting from these centrosome defects. Combinations of increases in number, size, abnormal structure and function characterize centrosome amplification. Centrosome amplification has been detected in a broad range of solid tumors and more recently in leukemia and lymphoma. Centrosome abnormalities have been shown to increase with increasing tumor grade and aggressiveness. Cells over-expressing pericentrin, a centrosome protein, mimic many of the cellular characteristics of a high-grade carcinoma. Therefore centrosome abnormalities are also a very good indicator of cancer progression. Both of these potential markers for carcinogenesis will be followed using flow cytometry-based assays. Pairs of mammary epithelial and fibroblast cells from the same individuals were used to better understand the etiology of cancer, and whether certain cell types respond differentially to radiation exposure. These studies complement our earlier studies using a normal hTERT immortalized skin fibroblast line, and help us identify unique signaling patterns specific to different cell types. By using two cell types from the same individual we will identify genotype influences in response. How track structure may effect the biological response will be considered by carefully choosing particles of various energies to obtain differing populations of cells hit to a greater or lesser proportion by the core particle or delta-rays to then investigate how phosphorylation profiles and surrogate cancer endpoints are affected by these components. The intent of this proposal is to provide a comprehensive picture of HZE effects from early to late phosphorylation phosphorylation profiles, and the relationships to various surrogate markers for carcinogenesis. Studies will be performed using novel flow cytometry-based assays, which have been optimized in our lab, to more rapidly and less subjectively monitor these endpoints. Materials & Methods: Cells were set up in flasks or chamber slides prior to irradiation, irradiated at the various doses and then fixed at multiple time points thereafter. A methanol and a combination paraformaldehyde and methanol fixation were used. Fixed slides were sent back from Brookhaven to our home institution in PBS to stain with the appropriate primary and secondary antibodies. Cells exposed in flasks were trypsinized and fixed at various times after exposure, and then sent back on ice in methanol. Aliquots of fixed cells were then stained for various phospho-proteins and analyzed via flow cytometry. For analysis at longer timepoints after exposure (weeks) cells were grown to confluence while at Brookhaven and then frozen down before leaving (~ 1 week post exposure). Cells were then thawed back at the home institution, and expanded to then analyze for telomere length changes and centrosome defects. Cells were fixed with a methanol fix for flow cytometry based analysis of centrosome dysfunction, and were hybridized to a telomere probe to allow detection of telomere length changes. Brief summary of past years research accomplishments: Normal hTERT immortalized fibroblasts were exposed to a very comprehensive range of carefully chosen high LET ions and energies, to help identify the relative contribution of delta rays and track structure, and then analyzed for phospho-kinetics of gamma H2AX, pSMC1 and pATF2. Fibroblasts were exposed in triplicate independent experiments to: Fe 300, 600, and 1000 MeV/u; Ti 300, 326, and 386 MeV/u; Si 93, 300, and 1000 MeV/u at timepoints between 0.5 and 24 h. Various comparisons for early and late timepoints were made between equal LET, energy or Z2/beta2 (roughly equivalent to the number and intensity of delta rays). These studies revealed Z2/beta2 to be a primary determinant in the phosphorylation response. In addition, residual levels of foci (24 h post exposure) correlated well with the predicted energy deposition per nucleosome. To determine whether phospho-kinetics are unique to the cell line assayed we also compared gamma H2AX, pSMC1, and pATF2 phosphorylation profiles in skin fibroblasts, mammary fibroblasts and genetically matched mammary epithelial cells following Fe 300 MeV/u, Ti 300 MeV/u, Si93 MeV/u, and Xray exposure. A final triplicate sample for some of the ions/energies was needed due to technical issues at previous runs and was obtained in the summer of 2012. These last pieces of phospho-signaling data will be analyzed on a single cell basis and compiled with our other data and incorporated into a drafted paper of this work. A general conclusion that can be made thus far is that cells originating from different tissues reveal different phospho-signaling kinetics, with different cell types exhibiting more similar kinetics when from the same tissue than same-cell types originating from different tissues. Skin fibroblasts revealed the greatest induction, with mammary epithelial and fibroblast cells showing a similar much dampened response in comparison. However both skin and mammary fibroblasts appear similar in that they both show a greater level of phospho-signaling than do epithelial cells when compared to their non-irradiated counter part. Conclusions from all phospho-kinetic studies thus far include that 1) high LET induces a greater induction of gamma H2AX and higher level of residual signaling as compared to equal doses of low LET; and 2) that at early timepoints ATM-mediated

phospho-inductions are much lower following all high LET ion exposures as compared to low LET, indicating that the ATM protein is hampered in fully activating down stream proteins following high LET irradiation as compared to low LET exposures, even at higher doses where all cells in the high LET exposed population should be hit.

In order to better understand the relationship between radiation exposure and carcinogenesis we have also analyzed how these various radiation qualities effect two surrogate cancer endpoints, telomere length and centrosome amplification. Flow cytometry-based telomere length assay, or flow FISH was used to determine how various qualities of radiation affected telomere lengths of cells differentially. In addition, the ability of different radiation qualities to induce centrosome aberrations was also studied and compared to phospho-signaling and telomere length changes in the same cells. A dose-dependent shortening of telomeres for both mammary epithelial and fibroblast cells following high LET exposures was observed, whereas low LET exposure resulted in only a small similar decrease following all doses up through 1Gy. Mammary epithelial cells were revealed to be more sensitive to radiation-induced telomere shortening than fibroblasts. In addition, differences between the cell types were also noted when telomere length changes where graphed based on fluence with all ions/energies except Fe 300 MeV/u, which surprisingly showed a nearly identical pattern for both cell types. Additional work was needed to optimize the flow-based assay to detect and analyze the percentage of centrosome aberrant cells. In addition studies were performed to determine the time dependence of centrosome aberrations following radiation, and we noted a significant increase in centrosome defects starting on day 9 after radiation exposure. We now believe we have optimized this assay and are in the process of completing staining, flow cytometry collection and single cell analysis for this endpoint. Thus far we have only analyzed experiments completed prior to this additional optimization, and see considerable variation in dosed sample results. It appears that there may be some suggestion of a greater level of centrosome aberrations with dose, but no clear evidence of a dose response. We have also compared fibroblast and epithelial cells for centrosome defects following X-ray exposure and found that only epithelial cells show an increase in centrosome defects with radiation exposure, whereas fibroblasts do not. Further studies will be done comparing centrosome defects in these cell types after high LET exposure. We have also looked at phospho-signaling in these cells which have been grown for weeks post exposure and see evidence of residual foci many weeks post exposure, in some cases even residual tracks appeared evident in cells, perhaps indicating delayed genomic instability and possibly increased levels of ROS in these previously irradiated cells. We are currently investigating how ROS levels may vary in cells after previous exposure to high or low LET and whether differences are noted based on cell types.

In summary the telomere, centrosome, and phospho-kinetic studies reveal that high LET is more effective than low LET in inducing telomere length shortening, but not centrosome defects and that increased phospho-signaling can be observed many weeks post exposure in radiation exposed cells. Centrosome studies need to be completed to then determine how this endpoint compares to the radiation quality dependent differences observed in telomere length as well as the residual foci observed at these late timepoints.

Bibliography Type: Description: (Last Updated: 11/01/2023)

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